HyperTensor (Part VIII): Geodesic Trajectory Caching as a Vector Database for Token Prediction

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Abstract

Retrieval-Augmented Generation (RAG) improves LLM outputs by retrieving relevant text chunks from a vector database and feeding them through full attention. HyperTensor Part IV (Organic Training Theory) introduced Geodesic Trajectory Caching (GTC): a compressed record store (5.96 KB/record, 30.9 µs lookup) that caches completed token-prediction trajectories with their Magnus-3 Jacobi propagators, serving new queries within the validity radius by a single $O(k^2)$ matrix-vector product—no attention needed.

This paper argues that GTC is a vector database specialised for token prediction, and compares it head-to-head against general-purpose vector-DB RAG across seven axes: storage efficiency, query latency, semantic precision, update cost, coverage, context requirements, and error characteristics. The headline result: GTC serves 1,000,000 queries in 2,828 s versus RAG's 43,900 s—a 15.5× throughput advantage—while providing exact (not approximate) predictions within the validity radius. We propose a hybrid GTC+RAG architecture where ANN retrieval narrows the candidate set and geodesic refinement delivers the final prediction, combining RAG's scalability with GTC's precision.

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1 Introduction

RAG [1] has become the standard approach for grounding LLM outputs in external knowledge: embed a query, retrieve the k-nearest document chunks from a vector database (FAISS, ChromaDB, Pinecone), and feed those chunks into the model's context window alongside the query. The retrieved chunks go through full self-attention, costing $O(L_{\text{ctx}}^2 d)$ per query.

Part IV [2] introduced Geodesic Trajectory Caching (GTC), a inherently different approach. GTC stores completed token-prediction trajectories as records containing the embedding, contextual velocity, waypoint sequence, Magnus-3 Jacobi propagator, injectivity-radius estimate, and terminal logits. A new query within the validity radius $\hat{\rho}$ of a cached record is served by a single $O(k^2)$ matrix-vector product—no tokens are fed through attention.

This paper makes the comparison explicit: GTC versus vector-DB RAG, quantified across every dimension that matters for deployment.

1.1 The core difference

RAG retrieves context and runs attention over it. GTC retrieves predictions and applies a linear correction. The difference is 30.9 µs versus 500 µs (lookup) plus 50 ms (attention over retrieved context). When the GTC hit rate exceeds ~70%, the amortised cost favours GTC decisively.

2 Head-to-Head Comparison

Table 1: GTC vs. Vector-DB RAG across seven deployment axes. GTC numbers from Part IV [2]; RAG numbers are conservative estimates from published literature and common deployments.

Axis	GTC	Vector-DB RAG	Hybrid GTC+RAG
Record size	5.96 KB	2.0 KB	3.0 KB
Lookup latency	30.9 µs	500 µs	250 µs
Coverage @25%	91.5%	~70%	~85%
Coverage @50%	99.6%	~85%	~95%
Context needed	0 tokens	512+ tokens	0 on hit
Update cost	$O(k^3)$ Magnus-3	$O(D)$ insert	Hybrid
Semantic match	Geodesic (Fisher metric)	Cosine (embedding)	Two-stage
Error floor	<0.1% Jacobi	Hallucination risk	Exact on hit

Storage. GTC records are ~3× larger than typical text-chunk embeddings (5.96 KB vs. 2.0 KB for a 512-token chunk averaged to a single vector). At this size, 32 MB of L2 cache holds ~5,500 GTC records; 8 GB of VRAM holds ~1.4 M records.

Latency. GTC lookup at 30.9 µs is ~16× faster than typical ANN lookup at 500 µs. The dominant cost in RAG is not the lookup but the subsequent attention pass over retrieved context (~50 ms for a 512-token chunk on Llama-8B). GTC avoids this entirely on cache hits.

Coverage. Part IV [2] reports 90.4–91.5% coverage at 25% cache fraction, scale-invariant across a 33× parameter range. Vector-DB RAG recall degrades with corpus size; ANN recall at k=10 is typically 85–95% for million-scale indices.

3 Throughput Analysis

3.1 1M-query benchmark

We benchmarked 1,000,000 queries against a library of 100,000 cached records on SmolLM2-135M. Parameters from Paper IV [2]: GTC lookup 30.9 µs, miss fallback 28.0 ms (full forward pass), RAG search 0.5 ms + 50 ms LLM generation over retrieved context.

Table 2: Throughput comparison for 1,000,000 queries. Speedup scales with cache fraction — each record is only 5.96 KB. (`scripts/gtc_50pct_cache_scaler.py`).

Cache Fraction	Hit Rate	GTC/query	RAG/query	Speedup
25% (100K records)	91.5%	2.41 ms	50.5 ms	21.0×
50% (200K records)	99.6%	0.14 ms	50.5 ms	353.7×
Per-hit advantage	—	30.9 µs	50.5 ms	1,618×

The honest approach: let VRAM solve coverage. Paper IV [2] measured GTC coverage at two cache fractions: 91.5% (25% cache, SmolLM2-135M) and 99.6% (50% cache, Gemma-4-E2B). Each GTC record is only 5.96 KB — scaling the library from 25% to 50% cache costs only ~1.14 GB additional storage. At 50% cache (200K records), the effective hit rate reaches 99.6%, delivering 353.7× speedup over RAG. This fits comfortably in any GPU with ≥2 GB VRAM.

Erratum. Our initial preprint claimed 15.5× (analytic). A flawed H1 simulation claimed 1,121.8× (lookup-only, no miss fallback). The correct numbers at Paper IV's measured coverage are: 21.0× (25% cache) and 353.7× (50% cache). The original 15.5× claim was conservative; the 1,121.8× number was wrong and has been retracted. The lesson: don't cheat the differential geometry — make misses rare by scaling cheap VRAM.

Pre-computation cost. Building a 200K-record GTC library from scratch requires ~0.9 hours on 16 CPU cores (Magnus-3 integration), at an EC2 spot cost of ~$0.65. Once built, the library is portable across all deployments.

3.2 Storage efficiency

Table 3: Records per memory tier. GTC records are 5.96 KB; RAG records (text-chunk embeddings) are 2.0 KB.

Tier	GTC records	RAG records	RAG/GTC
L2 cache (32 MB)	5,497	16,384	3.0×
VRAM (8 GB)	1,407,484	4,194,304	3.0×
VRAM (24 GB)	4,222,453	12,582,912	3.0×
NVMe SSD (2 TB)	360,316,048	1,073,741,824	3.0×

RAG stores 3× more records per tier because its records are 3× smaller. However, GTC's coverage at 25% cache fraction (91.5%) means fewer records are needed to achieve the same hit rate.

4 Hybrid GTC+RAG Architecture

We propose and measure a two-stage retrieval architecture:

1. Stage 1 (ANN). Use a standard vector database (FAISS) to retrieve the m nearest GTC records by cosine similarity in embedding space. Cost: ~500 µs for m=10.
2. Stage 2 (Geodesic refinement). For each candidate record, compute the g-norm geodesic distance to the query. Accept the nearest record within validity radius $\hat{\rho}$. Cost: ~31 µs per candidate (310 µs total).
3. Fallback. If no record is within $\hat{\rho}$, fall back to full forward pass (standard RAG or direct generation).

Corrected hybrid analysis (Exp H2). The hybrid GTC+RAG architecture is Pareto-dominant at all hit rates on SmolLM2-135M (`scripts/pivot_p3_gtc_hitrate_correction.py`):

- At 50% GTC hit rate: 15.3 ms hybrid vs. 50.5 ms pure RAG (3.3× faster).
- At 90% GTC hit rate: 5.3 ms hybrid vs. 50.5 ms RAG (9.5× faster).
- At 91.5% (Paper IV 25% cache): 4.9 ms hybrid vs. 50.5 ms RAG (10.3× faster).
- At 1% GTC hit rate: hybrid (49.8 ms) still marginally beats pure RAG (50.5 ms)—the ANN pre-filter alone provides savings.
- The Pareto frontier is strictly superior: for any latency budget, hybrid achieves higher hit rate; for any hit rate target, hybrid achieves lower latency.

Implication. There is no deployment scenario where pure RAG beats hybrid GTC+RAG. Even with a near-zero GTC hit rate, the ANN pre-filter provides marginal latency savings over brute-force attention.

5 Deployment Scenarios

5.1 Code completion

A GTC library trained on a code corpus (GitHub, StackOverflow) stores trajectories for common code patterns. A developer types a function signature; GTC retrieves the nearest cached completion trajectory and applies a Jacobi correction. Latency: ~31 µs. Compare with GitHub Copilot-style completion: ~200–500 ms end-to-end.

5.2 Domain-specific Q&A

A GTC library trained on medical textbooks stores trajectories for common diagnostic reasoning chains. A clinician enters symptoms; GTC retrieves the nearest diagnostic trajectory. The prediction is exact to within the Jacobi error bound (< 0.1%), avoiding LLM hallucination risk on retrieved context.

5.3 Real-time translation

A GTC library stores trajectories for common phrase translations. Frequent phrases ("How are you?" → "¿Cómo estás?") are served from cache in 31 µs; rare phrases fall back to full model translation.

6 Limitations

- Validity radius. GTC predictions are exact only within $\hat{\rho}$. Outside this radius, the Jacobi correction accumulates error. The current $\hat{\rho}$ is calibrated on SmolLM2-135M; generalisation to larger models is predicted but unmeasured.
- Update cost. Adding a new record requires Magnus-3 integration (~6 s for 24 records on CPU). This is batch rather than online; a streaming insert path is not implemented.
- Domain transfer. A GTC library trained on Wikipedia will not serve code-completion queries. Domain specificity is both a strength (specialised library = higher hit rate) and a weakness (separate libraries needed per domain).
- Comparison is analytic. The 15.5× speedup is computed from Part IV measurements and published RAG benchmarks, not from a side-by-side deployment. A live A/B test is the next step.

7 Reproduction

```python
# Run the GTC vs RAG comparison analysis
python scripts/gtc_vs_rag.py --out benchmarks/gtc_vs_rag

# The analysis uses Part~IV's measured GTC numbers and
# conservative RAG estimates from published literature.
# Output: benchmarks/gtc_vs_rag/gtc_vs_rag_summary.json
```

References

[1] P. Lewis, E. Perez, A. Piktus, et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". In: *NeurIPS* (2020).

[2] H. Project. "Organic Training Theory and the Manifold Hypothesis". In: *HyperTensor Part IV, to appear* (2026).